

AUTHORED BY: ARPIT RAJ, LNMIIT JAIPUR

Database Basics & Management

A database is actually a logical container that organizes and stores related objects.

These objects include:

Tables

Views

Indexes

Stored Procedures

Functions

Triggers

Events

Constraints

, there are multiple databases Reasons:

Better organization

Security

Independent backups

Easier maintenance

Scalability

Where Does MySQL Store a Database?

This is a favorite interview question.

When you execute

```
CREATE DATABASE College;
```

MySQL doesn't simply remember the name.

Internally it creates a directory on disk.

For example (Windows):

```
C:\ProgramData
  MySQL
    MySQL Server 8.0
      Data
        College\
```

Inside that directory, MySQL stores metadata and the files corresponding to the database objects. With modern MySQL versions using the InnoDB storage engine, table data is managed differently than older versions, but conceptually the database has its own directory.

2.2 CREATE DATABASE

The simplest syntax is

```
CREATE DATABASE College;
```

After execution,

MySQL creates `College`

and the database becomes available.

```
CREATE DATABASE College;
```

MySQL performs several checks.

Step 1

Checks whether `College` already exists.

Step 2

Checks whether the current user has permission.

Step 3

Allocates metadata.

Step 4

Creates the database directory and initializes the metadata needed to manage objects within it.

Returns Query OK

To avoid an error if the database already exists, use:

```
CREATE DATABASE IF NOT EXISTS College;
```

Now MySQL behaves like this:

If `College` doesn't exist



Create it.

If it already exists



Do nothing.

No error.

Is **College** the same as **college**

Answer:

It depends.

On Windows,

usually yes.

On Linux,

Linux filesystems are often case-sensitive

Selecting a Database

Creating a database doesn't automatically mean your subsequent commands will use it.

You must select it.

Syntax

```
USE College;
```

Now every table you create goes inside **College**

```
CREATE DATABASE College;
```

```
USE College;
```

```
CREATE TABLE Student(  
    ID INT,  
    Name VARCHAR(100)  
);
```

DROP DATABASE

Deletes an entire database.

Syntax

```
DROP DATABASE College;
```

This removes

Tables	Views
Indexes	Stored Procedures
Functions	Triggers
Data	

Everything inside the database is removed.

Internal Working

MySQL:

- Verifies permissions.
- Ensures no conflicting operations prevent the drop.
- Removes metadata.
- Deletes the database directory and associated files.

Afterward, recovery is generally only possible from backups.

```
DROP DATABASE IF EXISTS College;
```

If the database doesn't exist, MySQL simply skips the operation without throwing an error.